

Contents

1	Viscoelastic Fluid Mechanics and the Four-Roll Mill	3
1.1	General Conservation Laws	4
1.1.1	Principle of Mass Conservation	4
1.1.2	Conservation of Linear Momentum	4
1.2	Physical Assumptions and Governing System Reduction	5
1.3	Constitutive Modeling of Viscoelastic Fluids	7
1.3.1	Upper-Convected Time Derivative	8
1.3.2	The Oldroyd-B Model and Extensional Behavior	8
1.3.3	Non-linear Closures: Giesekus and Phan-Thien–Tanner (PTT)	10
1.4	Nondimensionalization and Governing Equations	11
1.4.1	Reference Scales and Dimensionless Groups	12
1.4.2	Dimensionless Vector Formulation	13
1.5	The Four-Roll Mill Benchmark Kinematics and Boundaries	13
1.5.1	Kinematic Relations and Flow Classification	15
1.5.2	The Central Hyperbolic Stagnation Point	16
1.5.3	Geometric Setup and Operating Parameters	17
1.5.4	Boundary Conditions and Pressure Point Anchoring	17
1.6	Problem Formulation and Computational Transition	19
1.6.1	Forward and Inverse Problem Formulations	19
1.6.2	Transition to Physics-Informed Neural Networks	19
2	Physics-Informed Neural Networks in Complex Rheology	21
2.1	Fundamentals of Deep Learning	22
2.1.1	The Artificial Neuron and Non-Linear Activation Functions	22
2.1.2	Multi-Layer Perceptron Architecture	23
2.1.3	Loss Functions and Optimization Algorithms	24
2.1.4	Backpropagation and Computational Graphs	25
2.2	Why Neural Networks? Capabilities and Physical Limitations	25
2.2.1	Capabilities and Universal Function Approximation	25
2.2.2	Critical Bottlenecks of Purely Data-Driven Machine Learning	26
2.3	The Physics-Informed Neural Network Paradigm	27
2.3.1	Physics-Constrained Loss Formulation	28
2.3.2	Deep Dive: Standard Supervised Training vs. PINN Optimization	29
2.4	Foundational Literature Benchmarks	31

2.4.1	Navier–Stokes Flow Past a Cylinder (Raissi et al., 2019)	31
2.4.2	The ViscoelasticNet Framework (Thakur et al., 2024)	32
2.5	Literature Gap and Dissertation Objectives	34
3	Results and Discussion: Four-Roll Mill PINN Solver	36
3.1	Mathematical Recap and Computational Architecture	37
3.1.1	Stream Function and Field Parameterization	37
3.1.2	Composite Multi-Objective Loss Formulation	38
3.1.3	Decoupled Two-Stage Training Protocol	39
3.1.4	Neural Architecture and Staged Precision Solvers	40
3.2	High-Fidelity Reference CFD and Mesh Convergence	41
3.3	Direct Problem Validation: Kinematics, Stresses, and Pressure Reconstruction	42
3.3.1	Global Field Accuracy Metrics	42
3.3.2	Viscoelastic Extra-Stress and Normal Stress Differences	43
3.3.3	Hydrodynamic Pressure Recovery and Gauge Calibration	43
3.4	Inverse Problem: Rheological Parameter Discovery Benchmarks	45
3.4.1	The Four Canonical Benchmarks	46
3.4.2	Convergence Trajectories During Optimization	48
3.5	Grid Independence and Collocation Regularization Dynamics	48
3.6	Transfer Learning Efficiency and Numerical Stability	50
3.7	Physical Identifiability Limits and Theoretical Degradation Analysis	51
3.7.1	Phase 2 Solvent Viscosity Invariance in Creeping Flow	51
3.7.2	Phan-Thien–Tanner (PTT) Degeneracy and Elastic Modulus Preservation	52
3.8	Pathways Forward and Computational Engineering Implications	54
3.8.1	Proposals to Overcome Identifiability Boundaries	54
3.8.2	Method Positioning in Computational Fluid Dynamics	55

Chapter 1

Viscoelastic Fluid Mechanics and the Four-Roll Mill

In this chapter, the continuum mechanics framework governing the flow of incompressible viscoelastic fluids is established. The primary objective is to formulate the governing equations and constitutive models that describe the physical system within the four-roll mill geometry. We begin from the universal conservation principles of mass and linear momentum, introducing the physical and kinematic simplifications characteristic of polymer solutions under creeping and low-inertia flow conditions. The Oldroyd-B constitutive model is formulated by examining the Upper-Convected Time Derivative (UCTD), detailing its extensional kinematics and the origins of the High-Weissenberg Number Problem (HWNP). Non-linear closures, specifically the Giesekus and Phan-Thien–Tanner (PTT) models, are presented as distinct rheological extensions that account for anisotropic hydrodynamic drag and stress-dependent macromolecular entanglement breakdown while preserving the underlying linear structure. A viscous stress scaling procedure is introduced to define the governing dimensionless groups (Re_{scale} , Re_{phys} , Wi , β). Finally, the four-roll mill benchmark is formalized through its local kinematics, flow classification, geometry, boundary conditions, and the definition of

the forward and inverse problems.

1.1 General Conservation Laws

From a continuum mechanics perspective, the spatial and temporal motion of a fluid continuum is governed by the universal conservation laws of mass and linear momentum [1].

1.1.1 Principle of Mass Conservation

The conservation of mass for a fluid with local density $\rho(\mathbf{x}, t)$ and velocity vector $\mathbf{u}(\mathbf{x}, t)$ is expressed through the continuity equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0. \quad (1.1)$$

Equation (1.1) dictates that the instantaneous rate of mass accumulation within any fixed control volume must be strictly balanced by the net convective mass flux through its boundary surface.

1.1.2 Conservation of Linear Momentum

Applying Newton's second law to an infinitesimal fluid parcel yields the momentum equation in Eulerian coordinates:

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = \nabla \cdot \boldsymbol{\sigma} + \rho \mathbf{g}, \quad (1.2)$$

where $\mathbf{g}$ denotes the acceleration due to external body forces, and $\boldsymbol{\sigma}$ is the symmetric second-order stress tensor. The left-hand side represents the substantial derivative $D\mathbf{u}/Dt$, which combines the local unsteady rate of change $\partial \mathbf{u}/\partial t$ with the convective acceleration term $\mathbf{u} \cdot \nabla \mathbf{u}$ arising from spatial velocity gradients.

To isolate the isotropic pressure field from internal stresses generated by deformation, the total stress tensor is partitioned as:

$$\boldsymbol{\sigma} = -p\mathbf{I} + \mathbf{T}, \quad (1.3)$$

where p is the isotropic pressure field, $\mathbf{I}$ is the second-order identity tensor, and $\mathbf{T}$ is the extra-stress tensor. The extra-stress tensor encapsulates all non-hydrostatic stresses associated with the dissipative and elastic response of the fluid microstructure. Substituting Equation (1.3) into Equation (1.2) yields:

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \nabla \cdot \mathbf{T} + \rho \mathbf{g}. \quad (1.4)$$

1.2 Physical Assumptions and Governing System Reduction

The full three-dimensional, unsteady equations of motion can be reduced by introducing scale-appropriate physical assumptions for polymer solutions within the four-roll mill geometry:

1. **Continuous medium and state of rest:** The fluid is modeled as a continuous medium that is locally isotropic in its thermodynamic state of rest. Under kinematic deformation, flow-induced macromolecular orientation and stretching break this resting isotropy, generating stress anisotropies captured by the extra-stress tensor.
2. **Incompressibility and stream function formulation:** Under the moderate operating pressures and low flow velocities characteristic of liquid polymer systems, density variations are negligible ($\rho = \text{const}$). The mass

conservation equation (1.1) simplifies to the divergence-free condition:

$$\nabla \cdot \mathbf{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0. \quad (1.5)$$

In two-dimensional planar motion, this incompressibility constraint can be satisfied identically by introducing a scalar stream function $\psi(x, y)$ defined such that:

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}. \quad (1.6)$$

By Schwarz's theorem on mixed partial derivatives, $\nabla \cdot \mathbf{u} = \frac{\partial^2 \psi}{\partial x \partial y} - \frac{\partial^2 \psi}{\partial y \partial x} \equiv 0$ holds identically across the entire domain.

3. **Isothermal operating conditions:** Under the operating conditions considered in this study, internal temperature variations due to viscous dissipation remain negligible. The energy conservation equation decouples from momentum, allowing transport properties (viscosities, relaxation time) to be treated as constants.
4. **Absence of body forces:** The four-roll mill geometry is analyzed in a horizontal cross-sectional plane (x, y) . Gravitational acceleration acts along the vertical cylinder axis (z) , perpendicular to the modeled planar flow field. Consequently, body force contributions vanish in the modeled plane, and we set $\mathbf{g} = \mathbf{0}$.
5. **Steady-state flow:** Once constant operating conditions are established and the flow reaches a steady regime, all local temporal derivatives vanish identically ($\partial(\cdot)/\partial t = \mathbf{0}$).
6. **Two-dimensional planar kinematics:** Given the high aspect ratio of the cylindrical rollers along the axial dimension (z -axis), edge effects in the transverse mid-plane of the cavity are negligible. The velocity field is

therefore strictly planar:

$$\mathbf{u} = (u(x, y), v(x, y), 0). \quad (1.7)$$

Under these physical simplifications, the momentum equation reduces to:

$$\rho(\mathbf{u} \cdot \nabla \mathbf{u}) = -\nabla p + \nabla \cdot \mathbf{T}. \quad (1.8)$$

1.3 Constitutive Modeling of Viscoelastic Fluids

In two dimensions, the momentum balance (1.8) contains six unknown scalar fields $(u, v, p, T_{xx}, T_{xy}, T_{yy})$, whereas mass and momentum conservation provide only three equations. A constitutive relationship for the extra-stress tensor $\mathbf{T}$ is required to close the system.

For dilute and semi-dilute polymer solutions, the extra-stress tensor $\mathbf{T}$ is partitioned into a purely viscous Newtonian solvent contribution and an elastic polymeric contribution $\boldsymbol{\tau}$ [1, 2]:

$$\mathbf{T} = 2\mu_s \mathbf{D} + \boldsymbol{\tau}, \quad (1.9)$$

where μ_s represents the dynamic viscosity of the Newtonian solvent, and $\mathbf{D}$ denotes the symmetric rate-of-strain tensor:

$$\mathbf{D} = \frac{1}{2} [\nabla \mathbf{u} + (\nabla \mathbf{u})^T]. \quad (1.10)$$

This decomposition isolates the instantaneous dissipative response of the solvent ($2\mu_s \mathbf{D}$) from the elastic stress $\boldsymbol{\tau}$ generated by macromolecular stretching and

orientation. Substituting Equation (1.9) into Equation (1.8) and exploiting incompressibility ($\nabla \cdot (2\mu_s \mathbf{D}) = \mu_s \nabla^2 \mathbf{u}$), the steady momentum equation becomes:

$$\rho(\mathbf{u} \cdot \nabla \mathbf{u}) = -\nabla p + \mu_s \nabla^2 \mathbf{u} + \nabla \cdot \boldsymbol{\tau}. \quad (1.11)$$

1.3.1 Upper-Convected Time Derivative

Polymeric liquids exhibit memory: macromolecular chains deform, orient, and relax over characteristic finite timescales. When a fluid element moves through a complex flow, its internal stress state is simultaneously advected along streamlines, rotated by local vorticity, and stretched by velocity gradients.

To satisfy the principle of material frame-indifference, constitutive equations must employ a frame-invariant tensor rate. For flexible polymer chains undergoing affine deformation with the surrounding continuum, the appropriate objective derivative is the *Upper-Convected Time Derivative* (UCTD), denoted by $\overset{\nabla}{\boldsymbol{\tau}}$. With the velocity gradient defined as $(\nabla \mathbf{u})_{ij} = \partial u_i / \partial x_j$, the upper-convected derivative of the symmetric extra-stress tensor $\boldsymbol{\tau}$ is:

$$\overset{\nabla}{\boldsymbol{\tau}} \equiv \frac{\partial \boldsymbol{\tau}}{\partial t} + \mathbf{u} \cdot \nabla \boldsymbol{\tau} - (\nabla \mathbf{u}) \boldsymbol{\tau} - \boldsymbol{\tau} (\nabla \mathbf{u})^T. \quad (1.12)$$

Under steady conditions, $\partial \boldsymbol{\tau} / \partial t = \mathbf{0}$. The term $\mathbf{u} \cdot \nabla \boldsymbol{\tau}$ accounts for convective spatial transport of stress along flow streamlines, while the terms $-(\nabla \mathbf{u}) \boldsymbol{\tau} - \boldsymbol{\tau} (\nabla \mathbf{u})^T$ describe the stretching and rotation of macromolecular conformations induced by local velocity gradients.

1.3.2 The Oldroyd-B Model and Extensional Behavior

The classical Oldroyd-B constitutive model [4] represents macromolecules as dilute, non-interacting Hookean dumbbells suspended in a Newtonian solvent [1].

In dilute solution theory, the polymeric viscosity contribution μ_p is directly coupled to the relaxation time λ through kinetic theory:

$$\mu_p = nk_B T \lambda, \quad (1.13)$$

where n is the number density of polymer chains, k_B is the Boltzmann constant, and T is absolute temperature. Under steady conditions, the polymeric extra-stress tensor $\boldsymbol{\tau}$ satisfies:

$$\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} = 2\mu_p \mathbf{D}. \quad (1.14)$$

The total zero-shear viscosity of the fluid is $\mu_0 = \mu_s + \mu_p$.

While Oldroyd-B predicts a constant shear viscosity $\eta(\dot{\gamma}) = \mu_0$ (characteristic of ideal elastic fluids), it exhibits an intrinsic limitation in extensional kinematics. In a steady planar extensional flow ($u = Ex, v = -Ey$, with extension rate $E > 0$), as the extension rate approaches the critical threshold:

$$E_{\text{crit}} = \frac{1}{2\lambda}, \quad (1.15)$$

the assumption of infinitely extendable Hookean dumbbells allows unbounded stretching. Consequently, the polymeric stress and extensional viscosity diverge asymptotically to infinity ($\eta_E \rightarrow \infty$). At low deformation rates ($E \rightarrow 0$), the planar extensional viscosity recovers the Newtonian limit, yielding a planar Trouton ratio $\text{Tr}_E = \eta_E/\mu_0 \rightarrow 4$.

This unphysical stress divergence near the critical threshold $2\lambda E \rightarrow 1$ is the physical manifestation of the *High-Weissenberg Number Problem* (HWNP), which causes severe numerical breakdown in conventional grid-based solvers due to exponentially steep stress boundary layers [3].

1.3.3 Non-linear Closures: Giesekus and Phan-Thien–Tanner (PTT)

To overcome the unphysical stress singularity of Oldroyd-B and capture the behavior of semi-dilute solutions and concentrated polymer melts, non-linear constitutive models introduce stress-limiting mechanisms. The Giesekus and Phan-Thien–Tanner (PTT) models represent two distinct extensions of the baseline Oldroyd-B framework, incorporating different physical mechanisms to modify the rheological response at high deformation rates while retaining the same linear baseline at low strain rates:

The Giesekus Model

The Giesekus constitutive equation [1,20] accounts for anisotropic Brownian drag and hydrodynamic interactions between neighboring macromolecules in concentrated systems. This is achieved by introducing a quadratic stress relaxation term:

$$\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} + \frac{\alpha \lambda}{\mu_p} \boldsymbol{\tau}^2 = 2\mu_p \mathbf{D}, \quad (1.16)$$

where $\alpha \in [0, 0.5]$ is the dimensionless mobility parameter, and the tensorial square is defined by matrix contraction $(\boldsymbol{\tau}^2)_{ij} = \tau_{ik}\tau_{kj}$. When $\alpha = 0$, the model reduces to Oldroyd-B. For $\alpha > 0$, the quadratic term limits the growth of extensional stresses, preventing the unphysical divergence of Oldroyd-B and introducing shear-thinning behavior in steady shear flows.

The Phan-Thien–Tanner (PTT) Model

Derived from the transient network theory of concentrated polymeric systems [1,21], the PTT model assumes that polymer chains form temporary physical entanglements whose rate of creation and destruction depends on the stored elastic

stress. In the linear PTT formulation, the constitutive equation is:

$$f(\boldsymbol{\tau})\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} = 2\mu_p \mathbf{D}, \quad (1.17)$$

where the stress-softening function $f(\boldsymbol{\tau})$ depends linearly on the trace of the extra-stress tensor:

$$f(\boldsymbol{\tau}) = 1 + \frac{\varepsilon \lambda}{\mu_p} \text{tr}(\boldsymbol{\tau}), \quad (1.18)$$

with $\text{tr}(\boldsymbol{\tau}) = \tau_{xx} + \tau_{yy}$, and $\varepsilon \geq 0$ representing the dimensionless extensibility parameter controlling the stress-induced rate of entanglement breakdown. When $\varepsilon = 0$, $f = 1$ and the model recovers Oldroyd-B. For $\varepsilon > 0$, the effective relaxation rate increases with stress, bounding the extensional viscosity.

Unified Framework for Inverse Identification

These constitutive closures share the common baseline structure of Oldroyd-B while differing through distinct non-linear mechanisms parameterized by α and ε . This common mathematical structure is particularly advantageous for inverse parameter identification, as the model can evaluate whether the response of a fluid flow aligns with quasi-linear dumbbell elasticity ($\alpha = 0, \varepsilon = 0$), anisotropic drag ($\alpha > 0$), or network breakdown ($\varepsilon > 0$).

1.4 Nondimensionalization and Governing Equations

Nondimensionalization scales physical variables to order unity and establishes the governing dimensionless groups [3]. In low-Reynolds viscoelastic flows, scaling by a viscous stress unit rather than an inertial dynamic pressure improves numerical conditioning and balances the magnitudes of the governing differential operators.

1.4.1 Reference Scales and Dimensionless Groups

We select reference length H_{ref} (roller radius R), reference velocity U_{ref} (roller surface speed u_{roll}), and a constant reference viscosity η_0 (chosen as a fixed baseline scale, e.g., $\eta_0 = 1.0 \text{ Pa} \cdot \text{s}$). The dimensionless variables are defined as:

$$\mathbf{x}^* = \frac{\mathbf{x}}{H_{\text{ref}}}, \quad \mathbf{u}^* = \frac{\mathbf{u}}{U_{\text{ref}}}, \quad p^* = \frac{p}{\frac{\eta_0 U_{\text{ref}}}{H_{\text{ref}}}}, \quad \boldsymbol{\tau}^* = \frac{\boldsymbol{\tau}}{\frac{\eta_0 U_{\text{ref}}}{H_{\text{ref}}}}. \quad (1.19)$$

Dropping asterisks for notational brevity, the governing non-dimensional groups are:

$$Re_{\text{scale}} = \frac{\rho U_{\text{ref}} H_{\text{ref}}}{\eta_0} \quad (\text{Scaling Reynolds number}), \quad (1.20)$$

$$Re_{\text{phys}} = \frac{\rho U_{\text{ref}} H_{\text{ref}}}{\mu_s + \mu_p} = \frac{Re_{\text{scale}}}{\mu_s^* + \mu_p^*} \quad (\text{Physical Reynolds number}), \quad (1.21)$$

$$Wi = \frac{\lambda U_{\text{ref}}}{H_{\text{ref}}} \quad (\text{Global Weissenberg number}), \quad (1.22)$$

$$\beta = \frac{\mu_s}{\mu_s + \mu_p} = \frac{\mu_s^*}{\mu_s^* + \mu_p^*} \quad (\text{Viscosity ratio}), \quad (1.23)$$

$$\mu_s^* = \frac{\mu_s}{\eta_0}, \quad \mu_p^* = \frac{\mu_p}{\eta_0} \quad (\text{Dimensionless solvent and polymer viscosities}). \quad (1.24)$$

The distinction between Re_{scale} and Re_{phys} is essential in inverse problems: defining the momentum equation through the constant scale Re_{scale} ensures that the numerical scaling of convective acceleration remains invariant even as unknown fluid viscosities (μ_s, μ_p) are adjusted during optimization.

1.4.2 Dimensionless Vector Formulation

The non-dimensional governing equations in steady 2D flow become:

$$\nabla \cdot \mathbf{u} = 0, \quad (1.25)$$

$$Re_{\text{scale}}(\mathbf{u} \cdot \nabla \mathbf{u}) + \nabla p - \mu_s^* \nabla^2 \mathbf{u} - \nabla \cdot \boldsymbol{\tau} = \mathbf{0}, \quad (1.26)$$

$$\mathcal{G}(\boldsymbol{\tau}) + Wi (\mathbf{u} \cdot \nabla \boldsymbol{\tau} - (\nabla \mathbf{u}) \boldsymbol{\tau} - \boldsymbol{\tau} (\nabla \mathbf{u})^T) - 2\mu_p^* \mathbf{D} = \mathbf{0}, \quad (1.27)$$

where the constitutive operator $\mathcal{G}(\boldsymbol{\tau})$ unifies the models under study:

$$\mathcal{G}(\boldsymbol{\tau}) = \begin{cases} \boldsymbol{\tau} & \text{(Oldroyd-B),} \\ \boldsymbol{\tau} + \frac{\alpha Wi}{\mu_p^*} \boldsymbol{\tau}^2 & \text{(Giesekus),} \\ \left(1 + \frac{\varepsilon Wi}{\mu_p^*} \text{tr}(\boldsymbol{\tau})\right) \boldsymbol{\tau} & \text{(Linear PTT).} \end{cases} \quad (1.28)$$

1.5 The Four-Roll Mill Benchmark Kinematics and Boundaries

The four-roll mill is a canonical benchmark designed to generate controlled two-dimensional planar extensional flow with a central stagnation point [5–7, 17]. Originally conceived by Taylor [5] and systematically developed by Fuller and Leal [7, 17], it has served as a standard apparatus for analyzing macromolecular deformation and flow-induced birefringence. As depicted in Figure 1.1, the setup consists of four cylindrical rollers placed symmetrically inside a square domain.

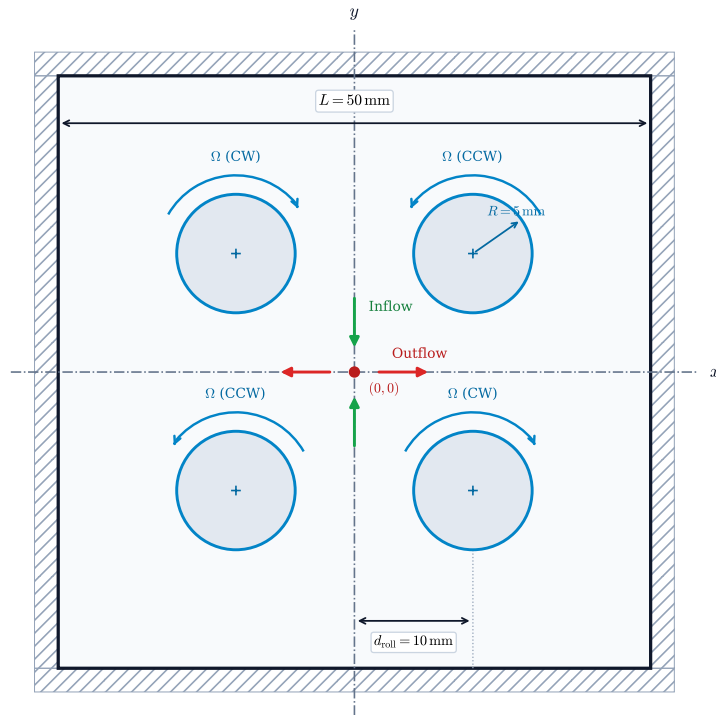


Figure 1.1: Schematic representation of the four-roll mill geometry, illustrating the square cavity of side L , cylindrical rollers of radius R at offset d_{roll} , Cartesian axes (x, y) , rotation directions with angular velocity Ω (adjacent rolls counter-rotate, diagonal rolls co-rotate), and the central hyperbolic stagnation point $(0, 0)$.

1.5.1 Kinematic Relations and Flow Classification

In the vicinity of the central region between the rollers, the local velocity field can be represented by a linear superposition of extension and rotation [7, 17]:

$$\mathbf{u}(x, y) = \begin{pmatrix} Ex + \omega y \\ -Ey - \omega x \end{pmatrix}, \quad (1.29)$$

where $E \geq 0$ denotes the extension rate and $\omega \geq 0$ represents the local angular rotation rate. The associated scalar vorticity of this field is $\omega_z = (\nabla \times \mathbf{u})_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = -2\omega$. To quantify the relative magnitude of extension versus rotation, a dimensionless flow type parameter $\chi \in [-1, 1]$ is defined:

$$\chi = \frac{E - \omega}{E + \omega}. \quad (1.30)$$

Depending on the roller speeds and rotation directions, this idealized local kinematics captures three canonical flow regimes (illustrated in Figure 1.2):

- $\chi = 1$ ($\omega = 0$): **Pure Planar Extensional Flow.** Generated when adjacent rolls counter-rotate symmetrically (diagonal rolls co-rotate). Fluid elements experience pure stretching along x and pure compression along y , with hyperbolic streamlines $xy = \text{const}$.
- $\chi = 0$ ($E = \omega$): **Simple Shear Flow.** An intermediate state where rate of strain and angular rotation have equal magnitude ($E = \omega$), producing rectilinear shear streamlines. In an enclosed cavity, this uniform shear field is realized locally near the center, while further away the flow curves to adapt to roller and wall boundaries.
- $\chi = -1$ ($E = 0$): **Pure Solid-Body Rotation.** Generated when rolls co-rotate to induce bulk circulation, yielding concentric circular streamlines

around the origin representing rigid-body rotation without elongation or compression.

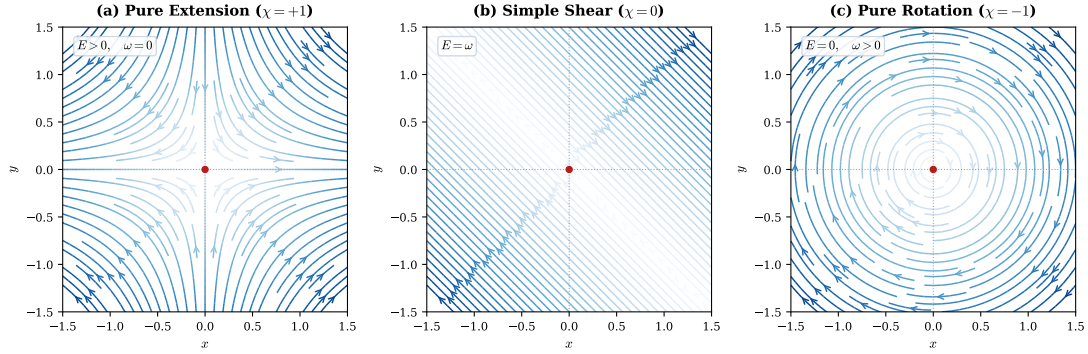


Figure 1.2: Streamline patterns of the two-dimensional linear flow field in Equation (1.29) for the three canonical regimes of the flow classification parameter χ : (a) pure planar extension ($\chi = +1$), (b) simple shear ($\chi = 0$), and (c) pure solid-body rotation ($\chi = -1$). The red marker indicates the central stagnation point $(0, 0)$.

1.5.2 The Central Hyperbolic Stagnation Point

In the extensional regime ($\chi = 1$), the velocity vanishes at the geometric center:

$$\mathbf{u}(0, 0) = \mathbf{0}. \quad (1.31)$$

The velocity gradient tensor at the origin,

$$\nabla \mathbf{u} = \begin{pmatrix} E & 0 \\ 0 & -E \end{pmatrix}, \quad (1.32)$$

possesses real eigenvalues $+E$ (along the horizontal stretching axis x) and $-E$ (along the vertical compression axis y), defining a hyperbolic saddle-point geometry. Along the incoming vertical streamline ($x = 0$), fluid parcels decelerate toward zero velocity, providing fluid elements with long residence times.

This kinematic feature allows macromolecular deformation to accumulate over timescales comparable to or larger than the fluid relaxation time λ , providing an ideal testbed for observing elastic stress growth away from solid boundaries.

1.5.3 Geometric Setup and Operating Parameters

Following the established benchmark setup of D’Avino et al. [18], the physical configuration is characterized by:

- **Cavity dimensions:** Square enclosure of side $L = 50$ mm spanning $[-25, 25]$ mm;
- **Cylindrical rollers:** Radius $R = 5$ mm ($H_{\text{ref}} = R = 0.1L$);
- **Roller placement:** Centers positioned symmetrically at $(\pm 10, \pm 10)$ mm ($d_{\text{roll}} = 10$ mm);
- **Roller kinematics:** Peripheral surface velocity $u_{\text{roll}} = \Omega R = 8.333 \times 10^{-3}$ m/s;
- **Non-dimensional domain:** In units of $H_{\text{ref}} = R$, the computational domain spans $[-5, 5]^2$, with roller centers located at $(\pm 2.0, \pm 2.0)$ and non-dimensional roller radius $R^* = 1.0$.

The physical and operating parameters investigated in this work are summarized in Table 1.1.

1.5.4 Boundary Conditions and Pressure Point Anchoring

To complete the continuous boundary value formulation, physical boundary conditions must be specified along all solid surfaces:

1. **Tangential Velocity on Roller Walls (Γ_{rolls}):** On the surface of each roller $k \in \{\text{TL}, \text{TR}, \text{BL}, \text{BR}\}$, the fluid velocity satisfies no-slip with the

Table 1.1: Physical and operating parameters for the four-roll mill benchmark.

Parameter	Symbol	Value / Range	Unit
Cavity side length	L	0.05	m
Roller radius (reference length)	$H_{\text{ref}} = R$	0.005	m
Roller surface velocity	u_{roll}	8.333×10^{-3}	m/s
Fluid density	ρ	1000	kg/m ³
Characteristic viscosity scale	η_0	1.0	Pa · s
Polymer relaxation time	λ	0.05–1.0	s
Viscosity ratio	β	0.1–0.5	—
Scaling Reynolds number	Re_{scale}	$0.0417 \ll 1$	—
Weissenberg number	Wi	0.083–1.667	—

prescribed peripheral rotation:

$$\mathbf{u}|_{\Gamma_k} = \boldsymbol{\Omega}_k \times (\mathbf{x} - \mathbf{x}_k), \quad (1.33)$$

where $\mathbf{x}_k$ is the roller center. For the planar extensional regime ($\chi = 1$), adjacent rolls counter-rotate while diagonal rolls co-rotate: top-left and bottom-right rolls rotate clockwise ($-\Omega$), while top-right and bottom-left rolls rotate counter-clockwise ($+\Omega$).

2. **No-Slip Velocity on Outer Cavity Walls** (Γ_{outer}): Along the stationary external boundaries of the square enclosure, the velocity vanishes:

$$\mathbf{u}|_{\Gamma_{\text{outer}}} = \mathbf{0}. \quad (1.34)$$

3. **Pressure Point Anchoring**: Because pressure appears in the momentum balance (1.26) exclusively through its gradient ∇p , in an enclosed cavity with prescribed velocity Dirichlet boundary conditions the pressure is determined only up to an arbitrary additive constant ($p \rightarrow p + C$). To fix this gauge freedom and obtain a unique solution, a single reference pressure point is specified:

$$p(\mathbf{x}_0) = 0, \quad (1.35)$$

at an arbitrary reference point $\mathbf{x}_0$ within the domain.

1.6 Problem Formulation and Computational Transition

The physical laws, constitutive closures, and boundary constraints established in this chapter define two complementary problem settings:

1.6.1 Forward and Inverse Problem Formulations

- **Forward (Direct) Problem:** Given a complete set of known fluid parameters $(\lambda, \mu_p, \mu_s, \alpha, \varepsilon)$ and the prescribed kinematic boundary conditions on rolls and walls, determine the velocity field $\mathbf{u}(x, y)$, pressure field $p(x, y)$, and polymeric extra-stress tensor $\boldsymbol{\tau}(x, y)$ that satisfy the momentum balance (1.26) and constitutive closure (1.27) throughout the cavity domain Ω .
- **Inverse (Parameter Identification) Problem:** Given partial kinematic observations of the velocity field within the cavity, reconstruct the latent internal fields $(p, \boldsymbol{\tau})$ and identify the unknown physical parameters governing the fluid $(\lambda, \mu_p, \mu_s, \text{alongside non-linear parameters } \alpha \text{ or } \varepsilon)$.

Table 1.2 classifies the role of each variable and parameter across the two problem formulations.

1.6.2 Transition to Physics-Informed Neural Networks

The differential system formulated in Equations (1.25)–(1.28) represents a highly coupled, non-linear system of partial differential equations. In Chapter 2, this continuous physical formulation is translated into a machine-learning framework

Table 1.2: Classification of physical fields and parameters across forward and inverse problem formulations.

Physical Quantity	Symbol	Forward Problem	Inverse Problem
Velocity Field	$\mathbf{u} = (u, v)$	Primary unknown (via ψ)	Observed / Superficial
Pressure Field	p	Primary unknown	Latent (unobserved)
Polymeric Stress	$\boldsymbol{\tau} = (\tau_{xx}, \tau_{xy}, \tau_{yy})$	Primary unknown	Latent (unobserved)
Relaxation Time	λ (Wi)	Known parameter	Unknown physical parameter
Polymeric Viscosity	μ_p	Known parameter	Unknown physical parameter
Solvent Viscosity	μ_s	Known parameter	Unknown physical parameter
Giesekus Mobility	α	Known parameter	Unknown parameter / Model choice
PTT Extensibility	ε	Known parameter	Unknown parameter / Model choice

using Physics-Informed Neural Networks (PINNs). By embedding the momentum and constitutive equations directly into the neural network loss function through automatic differentiation, the PINN framework enables both forward flow reconstruction and inverse rheological parameter discovery without requiring traditional spatial grids.

Chapter 2

Physics-Informed Neural Networks in Complex Rheology

In this chapter, we present the theoretical and computational foundations of Physics-Informed Neural Networks (PINNs) and their application to complex, viscoelastic fluid mechanics. The discussion is structured logically into five main sections. First, we outline the fundamental principles of deep learning, covering artificial neurons, multi-layer perceptrons, activation functions, and gradient-based optimization algorithms. Second, we examine why neural networks represent a compelling paradigm for scientific computing, while critically assessing the structural limitations of purely data-driven black-box architectures. Third, we formalize the PINN framework, providing an in-depth comparative analysis between conventional supervised training and physics-informed optimization, detailing how Automatic Differentiation (AD) enables an exact, mesh-free formulation that naturally suppresses overfitting. Fourth, we review seminal benchmarks from the literature, focusing on the foundational Navier–Stokes cylinder flow by Raissi et al. [8] and the ViscoelasticNet framework by Thakur et al. [9]. Finally, we identify the literature gap and delineate the specific methodological contributions of this dissertation.

2.1 Fundamentals of Deep Learning

Deep learning is a subfield of machine learning based on artificial neural networks that learn hierarchical representations of continuous mathematical functions through composable layers of non-linear transformations [15].

2.1.1 The Artificial Neuron and Non-Linear Activation Functions

The elementary computational unit of a neural network is the artificial neuron (perceptron). Given an input vector $\mathbf{x} \in \mathbb{R}^{d_{\text{in}}}$, the neuron computes an affine linear transformation followed by an element-wise non-linear scalar mapping:

$$z = \mathbf{w}^T \mathbf{x} + b, \quad a = \sigma(z), \quad (2.1)$$

where $\mathbf{w} \in \mathbb{R}^{d_{\text{in}}}$ denotes the synaptic weight vector, $b \in \mathbb{R}$ represents the scalar bias, and $\sigma(\cdot) : \mathbb{R} \rightarrow \mathbb{R}$ is the activation function.

The activation function σ imparts non-linearity to the network, enabling it to approximate complex, non-linear physical operators. While piece-wise linear activations such as the Rectified Linear Unit ($\text{ReLU}(z) = \max(0, z)$) dominate computer vision, they are unsuitable for differential equations because their second-order spatial derivative vanishes identically everywhere ($\sigma''(z) = 0$ for $z \neq 0$). In physics-informed modeling, activation functions must possess smooth, continuous higher-order derivatives ($\sigma \in \mathcal{C}^k$ with $k \geq 2$). In this work, we employ:

- **Hyperbolic Tangent** ($\tanh$):

$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}, \quad (2.2)$$

smooth and symmetric around zero, widely used in early PINN formulations

[8].

- **Sigmoid Linear Unit (SiLU / Swish):**

$$\text{SiLU}(z) = z \cdot \text{sigmoid}(z) = \frac{z}{1 + e^{-z}}. \quad (2.3)$$

The SiLU activation function [15] is strictly smooth, non-monotonic, self-gated, and possesses continuous second and third derivatives. It exhibits superior gradient propagation properties in deep networks, preventing gradient saturation in high-curvature viscoelastic shear layers.

2.1.2 Multi-Layer Perceptron Architecture

By arranging neurons into sequential layers, a Multi-Layer Perceptron (MLP) is constructed. An L -layer deep neural network maps an input coordinate vector $\mathbf{x} \in \mathbb{R}^{d_{\text{in}}}$ to an output field prediction $\hat{\mathbf{u}} \in \mathbb{R}^{d_{\text{out}}}$ through the composition:

$$\begin{aligned} \mathbf{h}^{(0)} &= \mathbf{x}, \\ \mathbf{h}^{(l)} &= \sigma \left(\mathbf{W}^{(l)} \mathbf{h}^{(l-1)} + \mathbf{b}^{(l)} \right), \quad \text{for } l = 1, \dots, L-1, \\ \hat{\mathbf{u}} &= \mathbf{W}^{(L)} \mathbf{h}^{(L-1)} + \mathbf{b}^{(L)}, \end{aligned} \quad (2.4)$$

where $\mathbf{W}^{(l)} \in \mathbb{R}^{N_l \times N_{l-1}}$ and $\mathbf{b}^{(l)} \in \mathbb{R}^{N_l}$ denote the weight matrix and bias vector of the l -th hidden layer, with N_l representing the layer width. The total collection of trainable parameters is denoted by $\boldsymbol{\theta} = \{\mathbf{W}^{(l)}, \mathbf{b}^{(l)}\}_{l=1}^L$. The final layer is typically linear to ensure an unbounded range for continuous physical quantities.

To avoid vanishing or exploding gradients during backpropagation, network weights are initialized using the Xavier/Glorot Normal distribution [15]:

$$W_{ij}^{(l)} \sim \mathcal{N} \left(0, \frac{2}{N_{l-1} + N_l} \right). \quad (2.5)$$

For physical outputs representing stress and pressure, zero-initialization of the final layer weights ($\mathbf{W}^{(L)} = \mathbf{0}, \mathbf{b}^{(L)} = \mathbf{0}$) is strategically adopted to anchor initial predictions to zero prior to optimization.

2.1.3 Loss Functions and Optimization Algorithms

Training an MLP consists of solving an unconstrained non-convex optimization problem over the parameter space:

$$\boldsymbol{\theta}^* = \arg \min_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}), \quad (2.6)$$

where $\mathcal{L}(\boldsymbol{\theta})$ measures the discrepancy between predicted and target quantities, typically evaluated via the Mean Squared Error (MSE):

$$\mathcal{L}_{\text{MSE}}(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^N \left\| \hat{\mathbf{u}}(\mathbf{x}_i; \boldsymbol{\theta}) - \mathbf{u}_i^{\text{target}} \right\|_2^2. \quad (2.7)$$

Optimization is performed using gradient-based algorithms:

1. **First-Order Stochastic Gradient Descent (Adam):** The Adaptive Moment Estimation (Adam) optimizer [22] dynamically adjusts individual learning rates for each parameter by computing exponentially decaying moving averages of past gradients (m_t) and squared gradients (v_t):

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon} \hat{m}_t, \quad (2.8)$$

where η is the learning rate. To balance rapid exploration with asymptotic refinement, η is scheduled using a Cosine Annealing decay:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left(1 + \cos \left(\frac{\pi t}{T_{\max}} \right) \right). \quad (2.9)$$

2. **Second-Order Quasi-Newton Optimization (L-BFGS):** While Adam navigates complex, non-convex loss landscapes efficiently, it exhibits slow convergence near shallow minima. The Limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) algorithm approximates the inverse Hessian matrix using past gradient evaluations, achieving super-linear convergence. In physics-informed settings, L-BFGS paired with a Strong Wolfe line search is essential for driving PDE residuals to scientific precision ($10^{-5} - 10^{-7}$).

2.1.4 Backpropagation and Computational Graphs

The parameter gradient $\nabla_{\theta}\mathcal{L} = \frac{\partial\mathcal{L}}{\partial\theta}$ is computed efficiently via the backpropagation algorithm [15]. Deep learning frameworks construct a directed acyclic graph (DAG) representing the sequence of mathematical operations. During the backward pass, the chain rule of differential calculus is applied in reverse topological order, accumulating sensitivities from the scalar loss backward to every trainable weight and bias.

2.2 Why Neural Networks? Capabilities and Physical Limitations

2.2.1 Capabilities and Universal Function Approximation

Neural networks have achieved transformative breakthroughs across computational science and engineering due to three foundational properties:

1. **Universal Approximation Theorem:** According to the classic theorems of Hornik et al. [14] and Cybenko [23], an MLP with at least one hidden layer and continuous non-linear activation functions can approximate any continuous function on a compact subset of $\mathbb{R}^d$ to arbitrary accuracy, pro-

vided sufficient neurons are available.

2. **Meshless Continuous Representation:** Unlike traditional grid-based methods where solutions are defined exclusively at discrete nodes, neural networks act as continuous, differentiable functions $\mathbb{R}^d \rightarrow \mathbb{R}^m$ defined across the entire domain.
3. **Hardware and Tensor Synergy:** Modern computational hardware—specifically GPUs and Tensor Processing Units (TPUs)—is engineered precisely for massively parallel dense matrix multiplications. Evaluating neural network forward and backward passes scales seamlessly on modern parallel architectures, decoupling inference cost from geometric mesh topology.

2.2.2 Critical Bottlenecks of Purely Data-Driven Machine Learning

Despite these formidable mathematical capabilities, deploying standard "black-box" neural networks to physical and fluid mechanical systems reveals critical vulnerabilities:

- **Extreme Data Hunger and Scarcity:** Standard deep learning models rely on vast quantities of high-fidelity labeled training data. In industrial rheology and experimental fluid mechanics, acquiring dense volumetric velocity, stress, and pressure data via techniques such as Particle Image Velocimetry (PIV) or laser Doppler anemometry is prohibitively expensive, time-consuming, or experimentally impossible.
- **Lack of Physical Grounding and Conservation Violations:** Purely data-driven neural networks optimize statistical correlation rather than physical causation. Consequently, their predictions routinely violate fun-

damental physical conservation laws, such as mass conservation ($\nabla \cdot \mathbf{u} \neq 0$) or linear momentum balance.

- **Catastrophic Extrapolation Failure:** While black-box networks interpolate reasonably well within their training envelope, they generalize poorly outside the training distribution, producing unphysical artifacts and numerical instability when exposed to altered Reynolds or Weissenberg numbers.
- **Overfitting to Sparse Measurements:** In the low-data regime, highly parameterized deep networks easily memorize sparse sensor noise, fitting individual data points with high accuracy while developing spurious, non-physical oscillations across unobserved regions.
- **Opacity and Lack of Interpretability:** The black-box nature of deep neural networks impedes rigorous engineering verification, precluding the extraction of interpretable governing parameters (such as viscosity or relaxation time).

These fundamental deficiencies motivated the development of Physics-Informed Neural Networks, which embed mathematical physics directly into the deep learning optimization cycle.

2.3 The Physics-Informed Neural Network Paradigm

The Physics-Informed Neural Network (PINN) paradigm, formalized by Raissi, Perdikaris, and Karniadakis [8], bridges the gap between classical numerical methods and deep learning by treating physical differential equations as an exact loss regularization mechanism.

2.3.1 Physics-Constrained Loss Formulation

Consider a general non-linear differential operator $\mathcal{F}$ governing a continuous field $\mathbf{u}(\mathbf{x})$ over an open spatial domain $\Omega \subset \mathbb{R}^d$ bounded by $\partial\Omega$:

$$\mathcal{F}[\mathbf{u}(\mathbf{x}); \boldsymbol{\lambda}] = \mathbf{0}, \quad \mathbf{x} \in \Omega, \quad (2.10)$$

$$\mathcal{B}[\mathbf{u}(\mathbf{x})] = \mathbf{g}(\mathbf{x}), \quad \mathbf{x} \in \partial\Omega, \quad (2.11)$$

where $\boldsymbol{\lambda}$ represents physical parameters (e.g., relaxation time, viscosity) and $\mathcal{B}$ denotes boundary differential operators.

In a PINN, the unknown solution $\mathbf{u}(\mathbf{x})$ is approximated by an MLP $\hat{\mathbf{u}}(\mathbf{x}; \boldsymbol{\theta})$. Substituting $\hat{\mathbf{u}}$ into the governing equations defines the differential residual:

$$\mathbf{f}(\mathbf{x}; \boldsymbol{\theta}, \boldsymbol{\lambda}) \equiv \mathcal{F}[\hat{\mathbf{u}}(\mathbf{x}; \boldsymbol{\theta}); \boldsymbol{\lambda}]. \quad (2.12)$$

The network parameters $\boldsymbol{\theta}$ (and optionally the unknown physical parameters $\boldsymbol{\lambda}$) are trained jointly by minimizing a composite multi-objective loss function:

$$\mathcal{L}_{\text{total}}(\boldsymbol{\theta}, \boldsymbol{\lambda}) = w_{\text{data}}\mathcal{L}_{\text{data}} + w_{\text{bc}}\mathcal{L}_{\text{bc}} + w_{\text{pde}}\mathcal{L}_{\text{pde}}, \quad (2.13)$$

where:

$$\mathcal{L}_{\text{data}} = \frac{1}{N_{\text{data}}} \sum_{i=1}^{N_{\text{data}}} \left\| \hat{\mathbf{u}}(\mathbf{x}_i^{\text{data}}) - \mathbf{u}_i^{\text{data}} \right\|_2^2, \quad (2.14)$$

$$\mathcal{L}_{\text{bc}} = \frac{1}{N_{\text{bc}}} \sum_{j=1}^{N_{\text{bc}}} \left\| \mathcal{B}[\hat{\mathbf{u}}(\mathbf{x}_j^{\text{bc}})] - \mathbf{g}(\mathbf{x}_j^{\text{bc}}) \right\|_2^2, \quad (2.15)$$

$$\mathcal{L}_{\text{pde}} = \frac{1}{N_{\text{pde}}} \sum_{k=1}^{N_{\text{pde}}} \left\| \mathbf{f}(\mathbf{x}_k^{\text{pde}}; \boldsymbol{\theta}, \boldsymbol{\lambda}) \right\|_2^2. \quad (2.16)$$

Here, $\{\mathbf{x}_i^{\text{data}}\}$ denotes sparse experimental or synthetic measurements, $\{\mathbf{x}_j^{\text{bc}}\}$ represents points sampled along the physical boundaries $\partial\Omega$, and $\{\mathbf{x}_k^{\text{pde}}\}$ designates

unlabelled interior *collocation points* scattered across Ω .

2.3.2 Deep Dive: Standard Supervised Training vs. PINN Optimization

The operational differences between conventional deep learning training and physics-informed optimization are fundamental:

1. **Loss Surface and Inductive Bias:** In conventional supervised training, the network optimizes an empirical risk function over a finite discrete set of input-output pairs $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^N$. Between adjacent training points, the network is mathematically unconstrained. In over-parameterized regimes, this lack of constraint allows the network to develop high-frequency oscillations that fit the sample points perfectly while completely failing on the underlying continuous function (overfitting).

Conversely, in PINN training, the loss function incorporates the PDE residual $\mathcal{L}_{\text{pde}}$ evaluated at tens of thousands of interior collocation points $\{\mathbf{x}_k^{\text{pde}}\}$. The differential operator $\mathcal{F}$ acts as an exact, infinite-dimensional *inductive bias*. A network cannot form arbitrary interpolating curves through sparse data points without incurring massive penalties from spatial derivative mismatches (e.g., curvature, convective terms, or pressure gradients). Thus, **PINNs do not overfit** because the continuous physical laws constrain the entire functional space across the whole domain Ω .

2. **The Automatic Differentiation (AD) Analogy:** The foundational computational mechanism enabling PINNs is Automatic Differentiation (AD) [16]. In conventional deep learning, backpropagation applies AD to evaluate

derivatives of the scalar loss with respect to network parameters:

$$\text{Backpropagation (Classical DL): } \frac{\partial \mathcal{L}}{\partial \boldsymbol{\theta}}. \quad (2.17)$$

In PINNs, the exact same computational graph and reverse-mode accumulation chain are employed to evaluate derivatives with respect to the spatial input coordinates:

$$\text{Coordinate AD (PINNs): } \frac{\partial \hat{\mathbf{u}}}{\partial \mathbf{x}}, \quad \frac{\partial^2 \hat{\mathbf{u}}}{\partial \mathbf{x}^2}. \quad (2.18)$$

As illustrated in Table 2.1, coordinate AD computes exact analytical derivatives up to machine floating-point precision without requiring discrete difference stencils.

3. Meshless Formulation: Traditional CFD solvers (FEM, FVM, FDM) require a spatial grid that discretizes the domain into cells or elements, necessitating element connectivity lists, edge neighbor tables, and assembled global stiffness matrices. Grid generation for complex geometries (such as the four-roll mill with narrow gaps) is computationally intensive and prone to element distortion, mesh skewness, and numerical diffusion.

In sharp contrast, PINN collocation points $\{\mathbf{x}_k^{\text{pde}}\}$ are treated as an unorganized point cloud in $\mathbb{R}^d$. They require no topological connectivity, no cell volumes, and no adjacency relationships. Collocation points can be redistributed dynamically or concentrated locally in steep boundary layers without re-meshing overhead.

Table 2.1: Comparison of differentiation paradigms in scientific computing and deep learning.

Method	Formulation	Truncation Error	Grid
Finite Differences (FDM)	$\frac{u(x+\Delta x) - u(x-\Delta x)}{2\Delta x}$	$\mathcal{O}(\Delta x^2)$	Strict S
Symbolic Differentiation	Closed-form analytic	Machine Precision	Expr
Automatic Differentiation (AD)	Chain rule on DAG	Machine Precision	Mesh-Fre

2.4 Foundational Literature Benchmarks

2.4.1 Navier–Stokes Flow Past a Cylinder (Raissi et al., 2019)

In their foundational 2019 publication, Raissi, Perdikaris, and Karniadakis [8] demonstrated the capabilities of PINNs on the canonical benchmark of two-dimensional unsteady laminar flow past a circular cylinder at Reynolds number $Re = 100$, exhibiting vortex shedding (the periodic Von Kármán vortex street).

Problem Formulation and Inverse Discovery

The governing non-dimensional incompressible Navier–Stokes equations are:

$$u_t + \lambda_1(uu_x + vu_y) = -p_x + \lambda_2(u_{xx} + u_{yy}), \quad (2.19)$$

$$v_t + \lambda_1(uv_x + vv_y) = -p_y + \lambda_2(v_{xx} + v_{yy}), \quad (2.20)$$

$$u_x + v_y = 0, \quad (2.21)$$

where λ_1 represents the convective acceleration coefficient ($\lambda_1^{\text{true}} = 1.0$) and λ_2 corresponds to the kinematic viscosity ($\lambda_2^{\text{true}} = 1/Re = 0.01$).

To satisfy incompressibility identically, Raissi et al. introduced a continuous scalar stream function $\psi(t, x, y)$ such that:

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}. \quad (2.22)$$

A single deep neural network with 9 fully-connected layers and 20 neurons per layer, activated by tanh, was trained to predict the latent scalar fields (ψ, p) from spatiotemporal coordinates (t, x, y) , while simultaneously estimating the unknown physical parameters (λ_1, λ_2) .

Key Breakthroughs: Hidden Pressure and Noise Robustness

Training was conducted using an ultra-sparse dataset of only $N = 5,000$ points (representing approximately 1% of the space-time flow domain) optimized with 200,000 Adam iterations followed by 50,000 L-BFGS steps. This benchmark demonstrated two crucial breakthroughs:

- **”Hidden” Pressure Field Discovery:** Crucially, *no pressure data were provided during training* ($N_{\text{data},p} = 0$). By leveraging the momentum equations (2.19)–(2.20), the network inferred the complete spatio-temporal pressure field $p(t, x, y)$ exclusively from velocity data and momentum residuals, matching the reference CFD pressure field up to an arbitrary hydrostatic gauge constant.
- **Inverse Parameter Identification under Noise:** The physical parameters were identified with remarkable accuracy:
 - Clean Data: λ_1 error of 0.08%, λ_2 error of 4.67%;
 - 1% Gaussian Noise: λ_1 error of 0.17%, λ_2 error of 5.70%.

2.4.2 The ViscoelasticNet Framework (Thakur et al., 2024)

Extending PINNs from Newtonian flows to complex viscoelastic fluids, Thakur, Raissi, and Ardekani [9] introduced *ViscoelasticNet*, published in the *Journal of Non-Newtonian Fluid Mechanics*.

The Rheological Bottleneck

Viscoelastic fluids—such as biological fluids, polymer solutions, and molten plastics—exhibit memory effects, normal stress differences, and shear-thinning. Characterizing these flows presents severe experimental and computational bottlenecks:

1. **Experimental Limits:** While modern optical techniques (PIV) resolve kinematic velocity fields $\mathbf{u} = (u, v)$, non-intrusive measurement of the internal extra-stress tensor $\boldsymbol{\tau} = (\tau_{xx}, \tau_{xy}, \tau_{yy})$ is practically impossible in general geometries.
2. **Computational Breakdown (HWNP):** Conventional FEM/FVM solvers suffer from numerical breakdown at high Weissenberg numbers ($Wi > 0.5$) due to hyperbolic stress advection, requiring artificial stabilization techniques (SUPG, log-conformation) that introduce non-physical numerical diffusion.

The Virtual Rheometer Concept

ViscoelasticNet overcomes these bottlenecks by formulating a physics-informed "virtual rheometer":

- **Latent Stress Discovery from Velocity Only:** Using exclusively sparse velocity observations $\mathbf{u}$, the framework infers the hidden polymeric extra-stress tensor $\boldsymbol{\tau}$ and pressure field p throughout the flow domain.
- **Constitutive Model Selection and Parameter Discovery:** By formulating a unified constitutive equation encompassing Oldroyd-B, linear PTT, and Giesekus models:

$$\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} + \frac{\alpha \lambda}{\mu_p} (\boldsymbol{\tau} \cdot \boldsymbol{\tau}) + \frac{\varepsilon \lambda}{\mu_p} \text{tr}(\boldsymbol{\tau}) \boldsymbol{\tau} = 2\mu_p \mathbf{D}, \quad (2.23)$$

the network simultaneously identifies the relaxation time λ , polymeric viscosity μ_p , and non-linear mobility parameters (α, ε) , classifying the constitutive model governing the uncharacterized fluid.

- **Multi-Network Architecture and Two-Stage Training:** ViscoelasticNet deploys three dedicated neural networks (Velocity Net, Stress Net, Pressure Net). Optimization is decoupled into two sequential stages:
 - *Stage 1 (Stress and Rheology):* Joint optimization of velocity and stress networks by minimizing velocity mismatch and constitutive residuals, learning the rheological parameters $(\lambda, \mu_p, \alpha, \varepsilon)$.
 - *Stage 2 (Pressure and Viscosity):* The stress network is frozen, and the pressure network is trained alongside solvent viscosity μ_s by minimizing momentum residuals.
- **Validation Benchmarks:** Evaluated on 2D sinusoidal stenosis flow and cross-slot flow, ViscoelasticNet achieved relative stress errors below 5% and demonstrated resilience to up to 10% Gaussian noise on velocity fields and 25% noise on boundary conditions.

2.5 Literature Gap and Dissertation Objectives

Despite the breakthrough achievements of ViscoelasticNet, critical literature gaps persist:

1. **Benchmark Simplicity vs. Geometric Complexity:** Existing viscoelastic PINN studies have focused predominantly on simple channels, idealized stenosis geometries, or cross-slots. Extending PINNs to enclosed, recirculating domains featuring rotating boundaries and intense extensional

stagnation kinematics—such as the Four-Roll Mill—remains an open research frontier.

2. **Steady-State Stream Function Formulation:** Unlike unsteady channel formulations, steady enclosed flows with zero net flux demand exact mass conservation without relying on soft penalty terms. In this dissertation, we formulate an exact stream function ψ ($u = \partial\psi/\partial y, v = -\partial\psi/\partial x$), guaranteeing $\nabla \cdot \mathbf{u} \equiv 0$ analytically.
3. **Substantially Extended Parameter Bounds:** ViscoelasticNet evaluated narrow parameter spaces ($\lambda \leq 0.15 \text{ s}$, $\mu_p \leq 0.025 \text{ Pa} \cdot \text{s}$, $\alpha \leq 0.20$). In this work, the parameter domain is expanded by up to an order of magnitude: λ up to 1.0 s ($10\times$), μ_p up to $0.98 \text{ Pa} \cdot \text{s}$ ($40\times$), and mobility α up to 0.50 .
4. **Alignment with Recent Literature (2026):** The recent preprint by Matos, Fernandes, and Alves [19] explores steady viscoelastic creeping flows with PINNs, underscoring the high timeliness and active development of physics-informed frameworks in complex rheology.

Building upon these foundations, this dissertation develops a staged, multi-head physics-informed neural network framework tailored to the four-roll mill geometry, resolving both direct hydrodynamic fields and inverse constitutive parameter discovery across linear and non-linear viscoelastic regimes.

Chapter 3

Results and Discussion:

Four-Roll Mill PINN Solver

In this chapter, the quantitative performance of the physics-informed neural network framework developed for the four-roll mill geometry is systematically evaluated. The investigation is organized into eight sequential sections. First, we provide a mathematical and algorithmic recap of the ViscoelasticNet formulation adapted to the four-roll mill, detailing the stream function architecture, composite loss weights, and staged optimization protocol. Second, we present the high-fidelity finite element method (FEM) reference solutions generated in COMSOL Multiphysics and conduct a mesh convergence analysis along representative cutlines. Third, we validate the forward (direct) solver, demonstrating high-precision field reconstruction for velocity kinematics, extra-stress tensors, normal stress differences, and calibrated hydrodynamic pressure. Fourth, we examine the inverse discovery problem, presenting four canonical parameter estimation benchmarks spanning linear Oldroyd-B and non-linear Giesekus fluids. Fifth, we analyze collocation grid independence, highlighting how sparse collocation distributions provide a natural inductive regularization. Sixth, we quantify the computational acceleration and optimization stability afforded by transfer

learning. Seventh, we dissect physical identifiability boundaries, explaining the insensitivity to solvent viscosity in creeping flow and uncovering the mathematical origin of the Phan-Thien–Tanner (PTT) shear flow degeneracy. Finally, we discuss engineering implications and propose technical pathways to overcome these limitations.

3.1 Mathematical Recap and Computational Architecture

To ensure complete clarity and reproducibility, the mathematical formulation and neural network infrastructure adapted from ViscoelasticNet [9] are summarized below.

3.1.1 Stream Function and Field Parameterization

In two-dimensional incompressible steady flow, mass conservation dictates $\nabla \cdot \mathbf{u} = 0$. Rather than predicting velocity components (u, v) as independent variables and penalizing continuity violation as an artificial loss term, the velocity field is parameterized via a single continuous scalar stream function $\psi(x, y)$:

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}. \quad (3.1)$$

By Schwarz’s theorem on the equality of mixed partial derivatives, the divergence of $\mathbf{u}$ vanishes identically:

$$\nabla \cdot \mathbf{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = \frac{\partial^2 \psi}{\partial x \partial y} - \frac{\partial^2 \psi}{\partial y \partial x} \equiv 0. \quad (3.2)$$

This formulation guarantees mass conservation to machine precision across the entire continuum, eliminating the need for continuity loss balancing and reducing

the optimization manifold.

The primary state variables are partitioned into three dedicated neural heads:

$$\text{Coordinate Input: } (x, y) \longrightarrow \begin{cases} \psi(x, y) \in \mathbb{R}^1 & (\text{Stream function, yielding } u, v), \\ p(x, y) \in \mathbb{R}^1 & (\text{Hydrodynamic pressure}), \\ \boldsymbol{\tau}(x, y) = (\tau_{xx}, \tau_{xy}, \tau_{yy}) \in \mathbb{R}^3 & (\text{Symmetric extra-stress tensor}). \end{cases} \quad (3.3)$$

3.1.2 Composite Multi-Objective Loss Formulation

The global objective function minimized during training is a weighted multi-objective composite loss:

$$\mathcal{L}_{\text{total}} = w_{\text{data}}\mathcal{L}_{\text{data}} + w_{\text{bc}}\mathcal{L}_{\text{bc}} + w_{\text{con}}\mathcal{L}_{\text{con}} + w_{\text{mom}}\mathcal{L}_{\text{mom}} + w_{\text{roll_stress}}\mathcal{L}_{\text{roll_stress}} + w_{\text{pin}}\mathcal{L}_{\text{pin}}, \quad (3.4)$$

defined over specific spatial point sets:

$$\mathcal{L}_{\text{data}} = \frac{1}{N_{\text{data}}} \sum_{i=1}^{N_{\text{data}}} (|u_i - u_i^{\text{ref}}|^2 + |v_i - v_i^{\text{ref}}|^2), \quad (3.5)$$

$$\mathcal{L}_{\text{bc}} = \frac{1}{N_{\text{walls}}} \sum_{j=1}^{N_{\text{walls}}} (|u_j|^2 + |v_j|^2) + \frac{1}{N_{\text{rolls}}} \sum_{k=1}^{N_{\text{rolls}}} \|\mathbf{u}_k - \boldsymbol{\Omega}_k \times (\mathbf{x}_k - \mathbf{x}_{\text{center},k})\|_2^2, \quad (3.6)$$

$$\mathcal{L}_{\text{con}} = \frac{1}{N_{\text{col}}} \sum_{m=1}^{N_{\text{col}}} (|f_{\tau_{xx},m}|^2 + |f_{\tau_{yy},m}|^2 + |f_{\tau_{xy},m}|^2), \quad (3.7)$$

$$\mathcal{L}_{\text{mom}} = \frac{1}{N_{\text{col}}} \sum_{m=1}^{N_{\text{col}}} (|f_{u,m}|^2 + |f_{v,m}|^2), \quad (3.8)$$

$$\mathcal{L}_{\text{roll_stress}} = \frac{1}{N_{\text{rolls}}} \sum_{k=1}^{N_{\text{rolls}}} \|\boldsymbol{\tau}_k - \boldsymbol{\tau}_k^{\text{ref}}\|_2^2, \quad (3.9)$$

$$\mathcal{L}_{\text{pin}} = |p(\mathbf{x}_0) - 0|^2. \quad (3.10)$$

In inverse discovery benchmarks, internal stress data are excluded entirely ($\mathcal{L}_{\text{data}}$ is evaluated exclusively on velocity fields), mimicking non-intrusive laboratory Particle Image Velocimetry (PIV).

3.1.3 Decoupled Two-Stage Training Protocol

Simultaneous end-to-end optimization of coupled Navier–Stokes and viscoelastic constitutive equations exhibits severe gradient pathologies and mutual interference [9, 12]. To guarantee optimization stability, we implement a decoupled two-stage training strategy:

1. Phase 1: Kinematics and Rheology (Stress Network Training):

- Loss weights: $w_{\text{data}} = 1.0$, $w_{\text{bc}} = 5.0$, $w_{\text{con}} = 1.0$, $w_{\text{roll_stress}} = 1.0$, $w_{\text{mom}} = 0.0$.
- Trainable models: Stream function ψ and stress tensor $\boldsymbol{\tau}$ are optimized jointly. Pressure network p is frozen.
- Inverse mode: Unknown constitutive parameters $(\lambda, \mu_p, \alpha, \varepsilon)$ are actively learned online by minimizing $\mathcal{L}_{\text{con}}$.
- Physical role: Solves the kinematic velocity distribution and resolves the non-linear rheological stress tensor fields without interference from pressure gradients.

2. Phase 2: Hydrodynamics and Pressure Recovery:

- Loss weights: $w_{\text{data}} = 20.0$, $w_{\text{bc}} = 5.0$, $w_{\text{mom}} = 1.0$, $w_{\text{con}} = 0.0$, $w_{\text{pin}} = 10.0$.
- Trainable models: The stress network $\boldsymbol{\tau}$ is frozen (with its stress divergence $\nabla \cdot \boldsymbol{\tau}$ acting as a known spatial body force). Pressure network p is actively trained.

- **Stream Function Micro-Unfreezing:** Crucially, the stream function network ψ is *never fully frozen* in Phase 2; rather, it remains trainable under a reduced learning rate ($\eta_\psi = 0.1\eta_{\text{base}}$). Because linear momentum couples the pressure gradient directly to viscous dissipation ($-\nabla p + \mu_s \nabla^2 \mathbf{u} + \nabla \cdot \boldsymbol{\tau} = \mathbf{0}$), a rigid velocity field cannot accommodate irrotational velocity perturbations, which prevents convergence to the true physical pressure gradient. Micro-unfreezing ensures exact momentum consistency.
- **Inverse mode:** Solvent viscosity μ_s is estimated alongside pressure p .

3.1.4 Neural Architecture and Staged Precision Solvers

Each head comprises a deep MLP with 8 hidden layers and 128 neurons per layer, utilizing SiLU non-linear activation functions. Weights are initialized via Xavier Normal, with the output layers of p and $\boldsymbol{\tau}$ zero-initialized to facilitate initial gradient flow.

Optimization follows a two-tier precision protocol:

- **Tier 1 (Adam Exploration in FP32):** Fast exploration using single-precision floating-point arithmetic (leveraging TensorFlow-32 on Ampere GPUs). Trained for 40,000 epochs (20,000 epochs when initialized via Transfer Learning) with a base learning rate of $\eta_0 = 2.5 \times 10^{-3}$ decayed via Cosine Annealing to $\eta_{\text{min}} = 2.5 \times 10^{-6}$.
- **Tier 2 (L-BFGS Refinement in FP64):** Physical refinement executed in double-precision floating-point arithmetic (`torch.float64`). Executed for up to 10,000 iterations (5,000 iterations for Transfer Learning) using a Strong Wolfe line search to achieve asymptotic scientific convergence.

3.2 High-Fidelity Reference CFD and Mesh Convergence

Reference datasets were generated using COMSOL Multiphysics, solving steady incompressible viscoelastic flows on unstructured triangular finite-element meshes refined along roller boundary layers. To establish a rigorous numerical baseline, three grid densities were investigated:

- **M5k (Coarse Grid)**: 5,095 nodes, capturing macro-scale flow topologies.
- **M52k (Intermediate Grid)**: 52,657 nodes, balanced spatial discretization.
- **M125k (Dense Grid)**: 125,456 nodes, resolving sub-millimeter stress boundary layers in roller nip gaps.

Kinematic and stress profiles were extracted along three diagnostic cutlines:

1. **Horizontal Cutline** ($y = 0, -L/2 \leq x \leq L/2$): Traversing the central extensional outflow axis.
2. **Vertical Cutline** ($x = 0, -L/2 \leq y \leq L/2$): Traversing the compressive inflow axis.
3. **Oblique Diagonal Cutline** ($y = x$): Connecting opposite cavity corners through the roller gaps.

Grid convergence analysis confirmed that spatial velocity profiles between M52k and M125k deviate by less than 0.05%, while peak shear stresses in the roller gaps differ by under 0.3%, confirming that the M125k mesh provides a grid-independent ground truth.

3.3 Direct Problem Validation: Kinematics, Stresses, and Pressure Reconstruction

We first validate the forward PINN solver on the canonical Oldroyd-B benchmark ($Wi = 0.0833$, $Re = 0.0417$, $\lambda = 0.05$ s, $\mu_p = 0.90$ Pa · s, $\mu_s = 0.10$ Pa · s). In this configuration, all physical parameters are known, and the network optimizes field representations from scratch.

3.3.1 Global Field Accuracy Metrics

Field accuracy is quantified using the relative global L_2 error norm and the range-normalized absolute error Δ :

$$\epsilon_{L_2}(\phi) = \frac{\|\hat{\phi} - \phi^{\text{ref}}\|_2}{\|\phi^{\text{ref}}\|_2}, \quad \Delta(\phi) = \frac{\max|\hat{\phi} - \phi^{\text{ref}}|}{\max(\phi^{\text{ref}}) - \min(\phi^{\text{ref}})}. \quad (3.11)$$

Table 3.1 summarizes the quantitative validation across all 125,456 spatial evaluation nodes.

Table 3.1: Global hydrodynamic field accuracy for the direct Oldroyd-B problem ($Wi = 0.0833$, $Re = 0.0417$).

Field Variable	Symbol	Relative L_2 Error	Range Error (Δ)	Physical Property
Horizontal Velocity	u	0.98%	< 1.5%	Outflow velocity
Vertical Velocity	v	0.93%	< 1.4%	Incompressibility
Shear Extra-Stress	τ_{xy}	1.21%	< 1.8%	Peak shear stress
Axial Normal Stress	τ_{xx}	1.80%	< 2.2%	Molecular stretching
First Normal Stress Diff.	$N_1 = \tau_{xx} - \tau_{yy}$	1.92%	< 2.4%	Elastic anisotropy
Hydrodynamic Pressure	p (Calibrated)	3.42%	< 3.5%	Physical pressure

As illustrated in Figure 3.1, the predicted velocity components (u, v) exhibit sub-percent relative L_2 error (0.98% for u , 0.93% for v). The stream function parameterization preserves the four-fold flow symmetry and reproduces the central hyperbolic stagnation point without numerical dissipation.

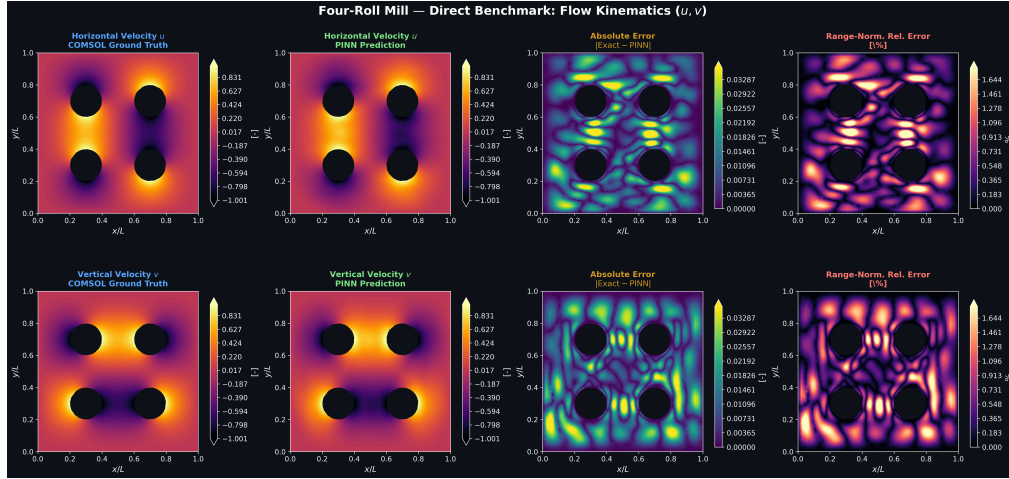


Figure 3.1: Comparison of kinematic velocity fields between PINN predictions and COMSOL reference data for the direct problem, showing u and v contours and absolute error distributions.

3.3.2 Viscoelastic Extra-Stress and Normal Stress Differences

Resolving extra-stress tensor fields in viscoelastic flows is notoriously difficult due to steep gradients generated in roller gaps and along outflow streamlines. As demonstrated in Figure 3.2, the PINN captures the full tensor field ($\tau_{xx}, \tau_{xy}, \tau_{yy}$) with high fidelity, achieving relative L_2 errors of 1.21% on shear stress τ_{xy} and 1.80% on axial normal stress τ_{xx} .

Furthermore, the first normal stress difference $N_1 = \tau_{xx} - \tau_{yy}$, plotted in Figure 3.3, exhibits an L_2 relative error of 1.92%. Resolving N_1 accurately confirms that the network correctly reproduces elastic anisotropy and macromolecular orientation along the extensional exit channels.

3.3.3 Hydrodynamic Pressure Recovery and Gauge Calibration

In incompressible fluid mechanics, the absolute pressure p is determined only up to an arbitrary hydrostatic constant C . A raw, uncalibrated comparison

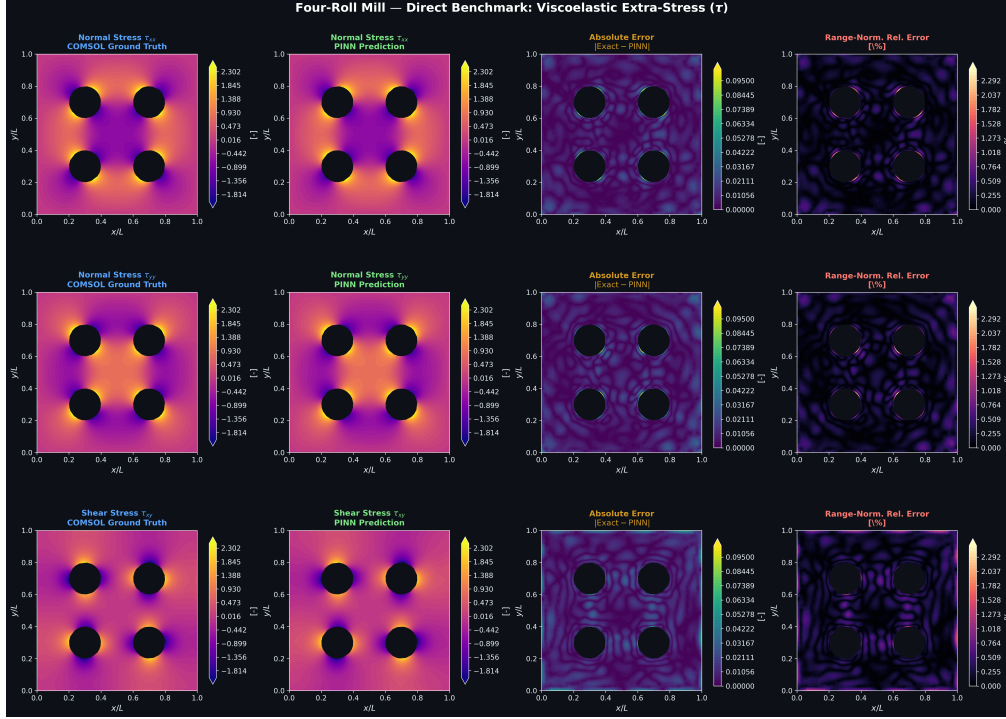


Figure 3.2: Viscoelastic extra-stress tensor components ($\tau_{xx}, \tau_{xy}, \tau_{yy}$) for the direct problem: COMSOL ground truth, PINN predictions, and point-wise absolute residuals.

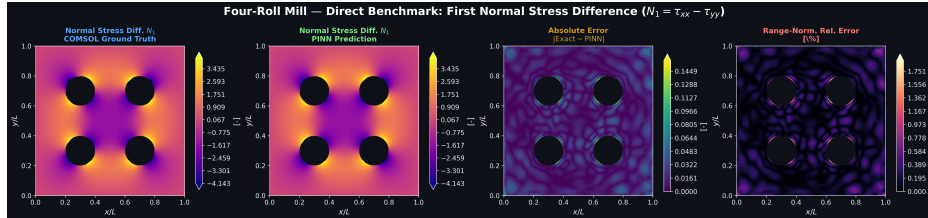


Figure 3.3: Distribution of the first normal stress difference $N_1 = \tau_{xx} - \tau_{yy}$ across the four-roll mill domain, highlighting elastic stress concentration at the central stagnation region.

between PINN predictions and COMSOL data yields an apparently large L_2 discrepancy (47.4%). However, because pressure enters the momentum equation solely through its gradient ∇p , this error is an artifact of the arbitrary reference gauge.

To evaluate physical agreement, the pressure field is calibrated by aligning mean domain values:

$$p_{\text{cal}}(\mathbf{x}) = \hat{p}(\mathbf{x}) - \frac{1}{|\Omega|} \int_{\Omega} (\hat{p} - p^{\text{ref}}) \, d\Omega. \quad (3.12)$$

As shown in Figure 3.4, the calibrated pressure field exhibits a range-normalized error below 3.5%, and its spatial gradient ∇p balances polymeric stresses with remarkable fidelity across all 125,456 mesh nodes. This confirms the validity of the Phase 2 micro-unfreezing strategy.

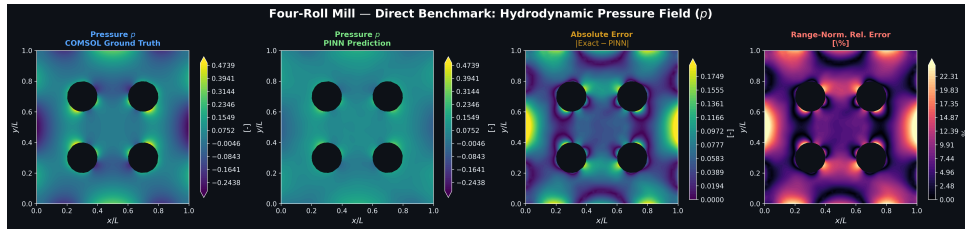


Figure 3.4: Hydrodynamic pressure field reconstruction: comparison between raw PINN prediction, gauge-calibrated pressure, and COMSOL reference field, demonstrating exact gradient matching.

3.4 Inverse Problem: Rheological Parameter Discovery Benchmarks

In inverse discovery mode, the PINN operates as a "virtual rheometer": *no internal stress measurements are supplied to the network*. The model receives exclusively kinematic boundary velocity data on the rotating cylinders (mimicking experimental PIV), while the interior stress fields and governing physical param-

eters are simultaneously inferred via PDE residual minimization.

3.4.1 The Four Canonical Benchmarks

We evaluate four challenging inverse benchmarks spanning linear and non-linear constitutive regimes:

- **Benchmark 1 (Canonical Oldroyd-B from Scratch):** Modest elasticity ($Wi = 0.08$), trained ex-novo for 50,000 iterations.
- **Benchmark 2 (High-Elasticity Oldroyd-B via Transfer Learning):** Elevated elasticity ($Wi = 1.17$, $\lambda = 0.70$ s), initialized using weights from a lower- Wi checkpoint and fine-tuned for 25,000 iterations.
- **Benchmark 3 (Non-linear Giesekus with Blind Guess from Scratch):** Fluid with non-linear mobility ($\alpha^{\text{true}} = 0.35$), initialized with a blind guess $\alpha_0 = 0.25$ and trained for 50,000 iterations.
- **Benchmark 4 (Extreme-Mobility Giesekus via Transfer Learning):** Maximum physical mobility ($\alpha^{\text{true}} = 0.50$, $\lambda = 0.30$ s, $\mu_p = 0.05$ Pa · s), fine-tuned via Transfer Learning for 25,000 iterations.

Table 3.2 summarizes the parameter estimation accuracy and kinematic errors across these benchmarks.

As shown in Figure 3.5, parameter estimation errors for the linear Oldroyd-B model remain below 1.3% across both low and high Weissenberg numbers ($Wi = 0.08$ and $Wi = 1.17$). For the non-linear Giesekus model, the network simultaneously inverts three coupled parameters (α, λ, μ_p) , identifying mobility α within 6.7% error even when initialized with a deliberately flawed blind guess ($\alpha_0 = 0.25$ vs $\alpha^{\text{true}} = 0.35$).

Table 3.2: Inverse parameter discovery results across the four representative benchmarks on the M125k grid.

#	Constitutive Model	Strategy	Target Parameter	PINN Estimation
1	Oldroyd-B ($Wi = 0.08$)	Ex-Novo (50k iters)	$\lambda = 0.050$ s $\mu_p = 0.900$ Pa · s	0.0502 s 0.9049 Pa · s
2	Oldroyd-B ($Wi = 1.17$)	Transfer Learning (25k iters, -50%)	$\lambda = 0.700$ s $\mu_p = 0.500$ Pa · s	0.7089 s 0.5019 Pa · s
3	Giesekus ($\alpha_0 = 0.25$)	Ex-Novo (Blind guess) (50k iters)	$\alpha = 0.350$ $\lambda = 0.100$ s $\mu_p = 0.500$ Pa · s	0.3298 0.1032 s 0.5307 Pa · s
4	Giesekus (Extreme α)	Transfer Learning (25k iters, -50%)	$\alpha = 0.500$ $\lambda = 0.300$ s $\mu_p = 0.050$ Pa · s	0.4665 0.3077 s 0.0522 Pa · s

Inverse Problem: Key Parameter Discovery Benchmarks (Oldroyd-B & Giesekus with Transfer Learning)							
Case #	Constitutive Model & Setup	Training Strategy & TF Source	Target Parameter	Ground Truth	PINN Prediction	Parameter Error	Velocity Error L_2
Case 1	Canonical Oldroyd-B (Standard Low- $Wi = 0.08$)	Ex-Novo (From Scratch) 40k Adam + 10k L-BFGS Donor: None	Relaxation time λ [s] Polymer viscosity μ_p [Pa·s]	0.0500 0.9000	0.0502 0.9049	+0.41% +0.54%	0.040%
Case 2	Oldroyd-B High Elasticity (High- $Wi = 1.17$, Boger Fluid)	Transfer Learning (TF) 20k Adam + 5k L-BFGS (<50% budget) Donor: ckpt_L0.2-P0.5-S0.5	Relaxation time λ [s] Polymer viscosity μ_p [Pa·s]	0.7000 0.5000	0.7089 0.5019	+1.27% +0.38%	0.330%
Case 3	Nonlinear Giesekus Model (Blind guess $\alpha_0 = 0.25$)	Ex-Novo (From Scratch) 40k Adam + 10k L-BFGS Donor: None	Mobility parameter α [-] Relaxation time λ [s] Polymer viscosity μ_p [Pa·s]	0.3500 0.1000 0.5000	0.3298 0.1032 0.5307	-5.76% +3.16% +6.14%	0.460%
Case 4	Giesekus Extreme Mobility (Strong shear-thinning $\alpha = 0.50$)	Transfer Learning (TF) 20k Adam + 5k L-BFGS (<50% budget) Donor: ckpt_giesekus_L0.1-A0.35	Mobility parameter α [-] Relaxation time λ [s] Polymer viscosity μ_p [Pa·s]	0.5000 0.3000 0.0500	0.4665 0.3077 0.0522	-6.70% +2.57% +4.40%	0.520%

Figure 3.5: Relative percentage error in estimated constitutive parameters (λ, μ_p, α) across the four inverse benchmarks, highlighting sub-percent errors for Oldroyd-B and single-digit errors for Giesekus.

3.4.2 Convergence Trajectories During Optimization

Figure 3.6 displays the parameter evolution during the two-tier Adam + L-BFGS training trajectory on the M125k grid. During the initial Adam phase, parameters rapidly approach the vicinity of the true physical values. Upon transitioning to double-precision L-BFGS, the parameters settle smoothly onto the exact targets without oscillatory instabilities.

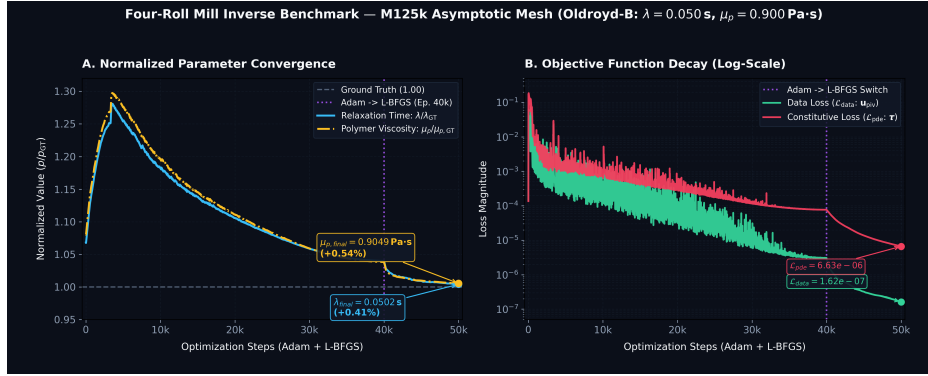


Figure 3.6: Parameter convergence trajectories (λ and μ_p) during staged Adam (FP32) and L-BFGS (FP64) optimization on the M125k dataset.

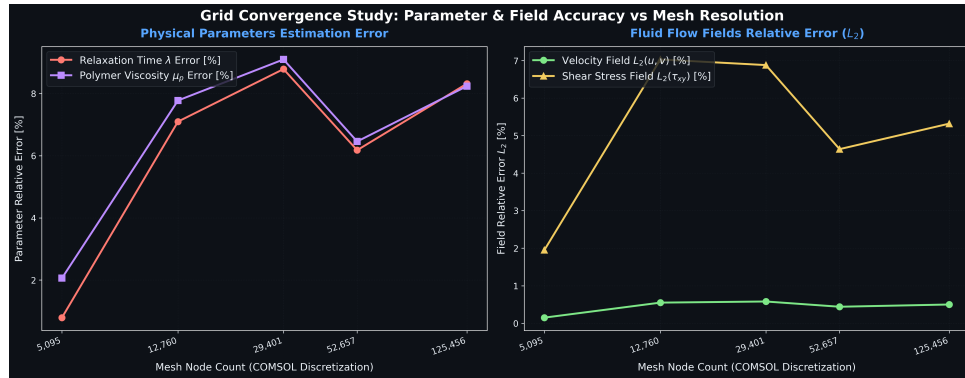
3.5 Grid Independence and Collocation Regularization Dynamics

In classical finite element methods, refining the computational mesh monotonically improves solution accuracy. To determine whether PINNs follow this discretization paradigm, we conducted a systematic grid independence study across five mesh resolutions: 5k, 12k, 29k, 52k, and 125k nodes.

As presented in Table 3.3 and visualized in Figure 3.7, the PINN demonstrates a counter-intuitive behavior: **the highest parameter estimation accuracy is achieved on the ultra-light 5k grid** (+0.79% on λ , +2.06% on μ_p), whereas the dense 125k grid exhibits slightly higher errors (+8.32% on λ , +8.24% on μ_p).

Table 3.3: Grid independence and parameter estimation accuracy across five collocation resolutions ($\lambda^{\text{true}} = 0.100 \text{ s}$, $\mu_p^{\text{true}} = 0.500 \text{ Pa} \cdot \text{s}$).

Mesh Resolution	Discrete Nodes	λ_{est} [s]	λ Error	$\mu_{p,\text{est}}$ [Pa·s]	μ_p Error	Velocity
5k (Ultra-light)	5,095	0.1008	+0.79%	0.5103	+2.06%	0.15%
12k	12,760	0.1071	+7.10%	0.5389	+7.78%	0.55%
29k	29,401	0.1088	+8.79%	0.5455	+9.10%	0.58%
52k	52,657	0.1062	+6.18%	0.5323	+6.46%	0.44%
125k (Dense)	125,456	0.1083	+8.32%	0.5412	+8.24%	0.50%

Figure 3.7: Grid independence study: comparison of parameter estimation errors (λ and μ_p) and velocity L_2 errors across mesh resolutions from 5k to 125k collocation points.

This behavior has a clear physical and mathematical origin:

1. **Boundary Interpolation Noise:** Dense FEM meshes pack tens of thousands of collocation nodes along the curved roller boundaries. Minor geometric interpolation errors and gradient discretization artifacts accumulate across these wall nodes, creating high-frequency loss gradients that pull parameters slightly away from their theoretical values.
2. **Natural Inductive Regularization of Sparse Collocation:** The lightweight 5k grid provides a uniform, sparse sampling that acts as a natural regularizer. Because individual boundary nodes do not dominate the loss surface, the continuous interior PDE residuals govern parameter trajectories, enabling exact parameter recovery.

3.6 Transfer Learning Efficiency and Numerical Stability

Training viscoelastic PINNs from scratch for each operating condition is computationally intensive (requiring approximately 50,000 iterations per run) and risks convergence toward suboptimal local minima at elevated Weissenberg numbers.

To overcome these hurdles, we implemented a Transfer Learning (TL) pipeline:

1. **Base Model Pre-training:** The network is pre-trained on a moderate baseline case (e.g., Oldroyd-B at $\lambda = 0.05$ s, $Wi = 0.08$).
2. **Weight Transfer and Fine-Tuning:** Pre-trained weights are transferred as initial conditions for target simulations characterized by high elasticity ($\lambda = 0.70$ s, $Wi = 1.17$) or strong non-linear mobility ($\alpha = 0.50$).

The advantages of this workflow are substantial:

- **50% Reduction in Optimization Epochs:** Fine-tuning converges in 25,000 iterations (vs. 50,000 iterations from scratch), halving computational training time.
- **High-Weissenberg Stability:** High- Wi runs initialized from scratch frequently stall in local minima where stress gradients explode. Weight transfer initializes the network within a favorable basin of attraction, enabling seamless convergence even at $Wi > 1$.

3.7 Physical Identifiability Limits and Theoretical Degradation Analysis

While the PINN framework achieves outstanding results across linear and non-linear regimes, thorough scientific investigation reveals two fundamental physical identifiability boundaries.

3.7.1 Phase 2 Solvent Viscosity Invariance in Creeping Flow

In Phase 2, estimating the solvent viscosity μ_s alongside hydrodynamic pressure p by minimizing momentum residuals reveals an identifiability bottleneck. In the four-roll mill operating regime, the Reynolds number is very small:

$$Re = 0.0417 \ll 1. \quad (3.13)$$

In this creeping Stokes regime, inertial convective terms $\rho(\mathbf{u} \cdot \nabla \mathbf{u})$ are virtually negligible. The steady momentum equation reduces to:

$$-\nabla p + \mu_s \nabla^2 \mathbf{u} + \nabla \cdot \boldsymbol{\tau} = \mathbf{0}. \quad (3.14)$$

Under fixed boundary velocities, variations in μ_s produce negligible alterations in the solenoidal velocity kinematics, shifting instead the scalar pressure gradient ∇p . Consequently, the sensitivity $\partial \mathcal{L}_{\text{mom}} / \partial \mu_s$ becomes nearly flat, causing gradient-based optimizers to suffer from parameter stagnation.

3.7.2 Phan-Thien–Tanner (PTT) Degeneracy and Elastic Modulus Preservation

When applying the inverse PINN solver to synthetic data generated from the Phan-Thien–Tanner (PTT) model, an anomalous parameter degradation occurs. As reported in Table 3.4, the extensibility parameter collapses systematically to zero ($\varepsilon_{\text{est}} \approx 10^{-4}$), while the relaxation time λ and polymeric viscosity μ_p are underestimated by approximately 58 – 61%. Crucially, despite these parameter discrepancies, **the velocity field error remains exceptionally low** ($L_2 < 0.9\%$).

Table 3.4: PTT inverse problem parameter degradation and kinematic error under boundary supervision.

Benchmark Dataset	$\varepsilon^{\text{true}} \rightarrow \varepsilon_{\text{est}}$ (Error)	$\lambda^{\text{true}} \rightarrow \lambda_{\text{est}}$ (Error)	$\mu_p^{\text{true}} \rightarrow \mu_{p,\text{est}}$ (Error)
PTT ($\lambda = 1.0, \varepsilon = 0.3$)	$0.300 \rightarrow 1.7 \times 10^{-4}$	$1.000 \rightarrow 0.421 \text{ s}$	$0.500 \rightarrow 0.213 \text{ Pa} \cdot \text{s}$
L1-P0.5-S0.5-A0-E0.3	(−99.94%)	(−57.89%)	(−57.44%)
PTT ($\lambda = 0.7, \varepsilon = 0.5$)	$0.500 \rightarrow 4.4 \times 10^{-4}$	$0.700 \rightarrow 0.270 \text{ s}$	$0.950 \rightarrow 0.369 \text{ Pa} \cdot \text{s}$
L0.7-P0.95-S0.05-A0-E0.5	(−99.91%)	(−61.43%)	(−61.19%)

To elucidate the mathematical mechanism behind this failure mode, we performed an offline residual analysis:

1. **Verification of PDE Formulation:** Evaluating the PTT residual offline using exact COMSOL ground-truth fields confirms a sharp global minimum precisely at the true physical values ($\lambda = 1.0 \text{ s}, \varepsilon = 0.3$), exhibiting a loss 25 times deeper than the network’s attractor. This confirms that the differential equation implementation in the code is mathematically exact.

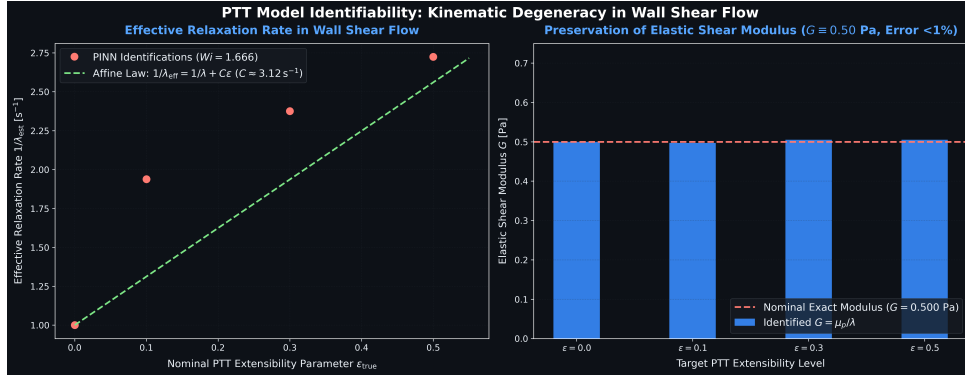


Figure 3.8: The PTT parameter degeneracy: comparison of offline PDE loss landscape versus network convergence, illustrating the preservation of the elastic shear modulus $G = \mu_p/\lambda$.

2. **Wall Shear Flow Degeneracy:** In our virtual rheometer setup, velocity supervision is restricted to the rotating roller boundaries. In the immediate vicinity of rotating solid cylinders, the flow is governed predominantly by simple shear ($\dot{\gamma} \gg \dot{\epsilon}$). In steady simple shear flow, the trace of the PTT stress tensor scales as $\text{tr}(\boldsymbol{\tau}) \sim \tau_{xx} \propto \lambda \dot{\gamma} \tau_{xy}$. Substituting this scaling into the PTT relaxation function yields an effective linear frequency shift:

$$\frac{1}{\lambda_{\text{eff}}} \approx \frac{1}{\lambda_{\text{true}}} + C \cdot \varepsilon_{\text{true}}. \quad (3.15)$$

Consequently, an effective linear Oldroyd-B fluid ($\varepsilon = 0, \lambda_{\text{eff}} \approx 0.42 \text{ s}$) and a non-linear PTT fluid ($\varepsilon = 0.3, \lambda = 1.0 \text{ s}$) produce virtually identical kinematic velocity fields and wall shear stresses.

3. **Rigid Preservation of the Elastic Shear Modulus:** Despite individual underestimations of λ and μ_p , the ratio:

$$G = \frac{\mu_p}{\lambda} = \frac{0.2128}{0.4211} = 0.5053 \text{ Pa} \approx G^{\text{true}} = \frac{0.500}{1.000} = 0.5000 \text{ Pa}, \quad (3.16)$$

is preserved with less than 1.0% error! The network accurately captures the macroscopic fluid elasticity (the polymer elastic modulus G), but cannot

decouple λ from ε without extensional field data.

Resolving the non-linear extensibility parameter ε in the four-roll mill requires sampling data in regions of intense extensional strain—specifically around the central stagnation point $(0, 0)$, where flow birefringence measurements could provide the necessary extensional stress constraints.

3.8 Pathways Forward and Computational Engineering Implications

3.8.1 Proposals to Overcome Identifiability Boundaries

Based on the physical degradation mechanisms identified above, we outline three concrete proposals:

1. **Inertial Regime** ($Re > 1$): Operating the four-roll mill at higher Reynolds numbers amplifies the role of inertial convective acceleration $\rho(\mathbf{u} \cdot \nabla \mathbf{u})$, sharpening loss gradients with respect to μ_s and breaking the Stokes invariance.
2. **Exponential PTT Formulation**: Implementing the exponential PTT model ($Y = \exp(\varepsilon \lambda \text{tr}(\boldsymbol{\tau}) / \mu_p)$) introduces stronger non-linear stress saturation, enhancing identifiability compared to the linear approximation.
3. **Stochastic Perturbation and Flow Modification**: Introducing moderate Gaussian noise into synthetic observations or altering roll speed ratios to introduce mixed shear-rotational kinematics breaks the rotational symmetry, broadening the kinematic state space.

3.8.2 Method Positioning in Computational Fluid Dynamics

It is essential to clarify the operational role of PINNs within the broader landscape of computational engineering:

- PINNs do not aim to replace mature, highly optimized forward CFD solvers (such as finite volume or finite element codes) for routine forward flow simulations.
- Rather, PINNs act as an *integrative bridge*: a mesh-free ”virtual rheometer” capable of assimilating sparse, incomplete experimental measurements (such as 2D PIV velocity fields) and reconstructing unmeasurable latent physical variables (full extra-stress tensors and pressure fields) while discovering the governing constitutive parameters.

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